

SWAMINARAYAN UNIVERSITY



FACULTY OF ENGINEERING

DEGREE ENGINEERING

DEPARTMENT OF IT

Introduction to Machine Learning (BE104705)

7th SEMESTER

ACADEMIC YEAR 2026-27

HOD

DEAN



**SWAMINARAYAN
UNIVERSITY**
THE RESERVOIR OF ALL KNOWLEDGE

FACULTY OF ENGINEERING
DIPLOMA ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY

Introduction to Machine Learning
ASSIGNMENTS

UNIT 1: Introduction to Machine Learning:

1. Explain the **Overview of Human Learning and Machine Learning**. Compare both with suitable examples.
2. Describe the **Types of Machine Learning**:
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learningwith examples.
3. Explain the **Applications of Machine Learning** in different fields such as healthcare, finance, education, e-commerce, and transportation.
4. Discuss the **Tools and Technologies used in Machine Learning**, including their features and uses.
5. Compare **Human Learning and Machine Learning** based on:
 - Learning process
 - Decision-making
 - Adaptability
 - Speed
 - Accuracy

UNIT 2: Preparing to Model:

1. Explain the **Machine Learning process (activities)** with a neat diagram.
2. Describe the **Types and Structures of Data** used in Machine Learning with suitable examples.
3. Explain **Data Quality** and discuss different **Data Remediation techniques**.
4. Explain the **Data Preprocessing** techniques used in Machine Learning, including:
 - Data Cleaning

- Data Transformation
 - Data Integration
 - Data Reduction
5. Explain the following feature selection techniques with suitable examples:
 - Dimensionality Reduction
 - Feature Subset Selection

UNIT 3: Modeling and Evaluation:

1. Explain the **process of selecting a Machine Learning model**. Differentiate between **Predictive** and **Descriptive** models with suitable examples.
2. Describe the **steps involved in training a supervised learning model** with a neat diagram.
3. Explain **Model Representation** and **Model Interpretability** with suitable examples.
4. Discuss the **different methods used to evaluate the performance of a Machine Learning model**.
5. Explain the **techniques used to improve the performance of a Machine Learning model**, such as:
 - Feature Selection
 - Hyperparameter Tuning
 - Cross-Validation
 - Handling Overfitting and Underfitting

UNIT 4: Basics of Feature Engineering:

1. Explain **Feature Engineering** and discuss its importance in Machine Learning.
2. Describe the **Feature Transformation** techniques:
 - Feature Construction
 - Feature Extraction
with suitable examples.
3. Explain **Feature Subset Selection** and discuss its role in improving Machine Learning models.
4. Discuss the **issues associated with high-dimensional data** and explain how they can be addressed.
5. Explain the **overall process of Feature Engineering**, including:
 - Key Drivers
 - Feature Measurement
 - Feature Selection
 - Feature Transformation

UNIT 5: Overview of Probability:

1. Explain the **Concepts of Probability** and discuss their importance in Machine Learning.
2. Describe **Random Variables** and explain the difference between **Discrete** and **Continuous Probability Distributions** with suitable examples.
3. Explain the **Central Limit Theorem (CLT)** and discuss its significance in statistical analysis and Machine Learning.
4. Explain the process of **Hypothesis Testing**, including:
 - Null Hypothesis (H_0)
 - Alternative Hypothesis (H_1)
 - Significance Level
 - Decision Making
5. Discuss the following statistical concepts used in Machine Learning:
 - Sampling Distributions
 - Multiple Random Variables
 - Monte Carlo Approximation

UNIT 6: Bayesian Concept Learning:

1. Explain the **importance of Bayesian methods** in Machine Learning with suitable examples.
2. State and explain **Bayes' Theorem**. Illustrate it with a numerical example.
3. Explain **Bayes' Theorem in Concept Learning** with a suitable example.
4. Describe the **Bayesian Belief Network (BBN)**, its components, working, advantages, and applications.
5. Compare **Bayesian Concept Learning** with traditional concept learning approaches.

UNIT 7: Supervised Learning:

1. Explain **Supervised Learning** with a neat diagram and suitable examples.
2. Describe the **Classification Model**, its learning steps, and applications.
3. Explain the **Classification Algorithms** used in Machine Learning, such as:
 - Decision Tree
 - Naïve Bayes
 - k-Nearest Neighbors (KNN)

- Support Vector Machine (SVM)
- 4. Explain **Regression** and discuss different **Regression Algorithms**, such as:
 - Linear Regression
 - Polynomial Regression
 - Logistic Regression
 - Decision Tree Regression
- 5. Compare **Classification** and **Regression** based on:
 - Output
 - Data Type
 - Objective
 - Algorithms
 - Applications

UNIT 8: Unsupervised Learning:

1. Explain **Unsupervised Learning** with a neat diagram and suitable examples.
2. Compare **Supervised Learning** and **Unsupervised Learning** based on:
 - Training Data
 - Output
 - Goal
 - Algorithms
 - Applications
3. Explain **Clustering** in detail, including its working, advantages, and applications.
4. Describe **Association Rules** and explain their importance in Machine Learning with suitable examples.
5. Discuss the **Applications of Unsupervised Learning** in different domains such as:
 - Customer Segmentation
 - Market Basket Analysis
 - Recommendation Systems
 - Anomaly Detection
 - Image Segmentation

UNIT 9: Neural Network:

1. Explain the **Introduction to Neural Networks** and compare **Biological Neurons** with **Artificial Neurons** using a suitable diagram.
2. Describe the **Types of Activation Functions**:
 - Sigmoid
 - ReLU

- Hyperbolic Tangent (Tanh)
with their advantages and disadvantages.

3. Explain the **Architecture of an Artificial Neural Network (ANN)**, including:

- Input Layer
 - Hidden Layer
 - Output Layer
- with a neat diagram.

4. Explain the **Learning Process in ANN** and describe the **Back Propagation Algorithm** with a suitable diagram.

5. What is **Deep Learning**? Explain its architecture, advantages, and real-world applications.